TER NOTEBOOK

towards Tate's Thesis

Contents

1 Riemann's zeta function

1.1 Riemann's idea I: analytical continuation of zeta function

As the core object in number theory, it is formally defined as a meromorphic function on a complex plane by Riemann in 1859, in his famous-9-page paper. **This section is the reconstruction of the zeta function along the Riemann's original ideas, it refers to pages 1 and 2 of the original paper.** .

$$\sum_{n\geq 1}\frac{1}{n^s}=\prod_p\frac{1}{1-p^{-s}} \tag{1}$$

This is Euler's product formula proved by Euler (around 1740), with the basic ideas of geometric series

$$\frac{1}{1-p^{-s}}=1+p^{-s}+p^{-2s}+\cdots$$

using unique factorization to build a basic correspondence between terms. The formal proof of the formula holds true in half-plane $\Re(s)>1$ is given by Dirichlet, it is important that the analytic formula makes sense and plays a core role in the analytic number theory. As a student of Dirichlet, Riemann knew the analytic results about the series exactly, and in his era, complex analysis was an advanced branch and analytic continuation was developed completely, hence Riemann tried to treat Euler's formula 1 as an analytic function instead of a formal identity:

$$\zeta(s)=\sum_{n\geq 1}\frac{1}{n^s},\quad \Re(s)>1$$

A similar object is the Gamma function, which was called "factorial function" in the age of Gauss and Euler, and Euler noticed an identity of integration:

$$n!=\int_0^\infty e^{-x}x^n dx,\quad n=1,2,3... \tag{2}$$

which motivates Gauss (1813) to introduce the notation

$$\Pi(s)=\int_0^\infty e^{-x}x^s dx,\quad \Re(s)>-1$$

By the mordern notation, $\Pi(s-1)=\Gamma(s)$. Before Riemann developed the zeta function, some important functional equations of the Gamma function were already known, especially the following results:

$$\Gamma(s+1)=s\Gamma(s) \tag{3}$$

$$\Gamma(s)\Gamma(1-s)=\frac{\pi}{\sin(\pi s)} \tag{4}$$

$$\Gamma(s)\Gamma\left(s+\frac{1}{2}\right)=2^{1-2s}\sqrt{\pi}\Gamma(2s) \tag{5}$$

the duplication formula was developed by Legendre (1809), many results about the Gamma function were developed by Euler and Gauss, Riemann surely knew these and he gave the analytic continuation of the zeta function by the following integration:

$$\int_0^\infty e^{-nx}x^{s-1}dx=\frac{\Gamma(s)}{n^s} \tag{6}$$

hence it is natural to consider the summation, it is not difficult to conclude that

$$\int_0^\infty \frac{x^{s-1}}{e^x-1}dx = \Gamma(s)\zeta(s), \quad \Re(s) > 1 \tag{7}$$

Then Riemann considered the contour integral

$$\int_{\mathcal{H}} \frac{(-z)^{s-1}}{e^z-1}dz$$

where the contour $\mathcal{H}$ can be drawn as following:

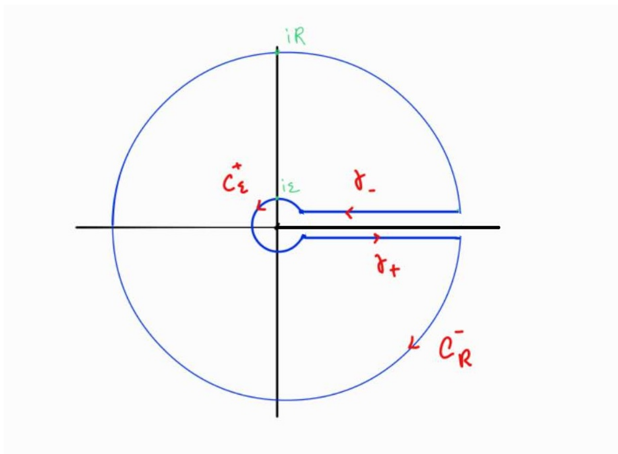

Figure 1: Henkel contour

This type of contour is called **the Henkel contour**, it was systematically studied by Hermann Hankel, who was the student of Riemann and infulenced by him greatly. For the sake of convience, we denote $\delta = \frac{\varepsilon}{\sqrt{2}}$ and $T = \sqrt{R^2-\varepsilon^2}$, and then we can write

$$\gamma_+(x) = x - i\delta, \quad (-\gamma_-)(x) = x + i\delta$$

for $x \in [\delta, T]$. Notice that $\mathcal{H}$ has negative orientation, and $f : z \mapsto \frac{(-z)^{s-1}}{e^z-1}$ has simple poles in the imaginary axis, so we choose ε sufficiently small such that $B(0,\varepsilon)$ only contains the pole at zero, then by the residual theorem we have

$$\int_{\mathcal{H}} \frac{(-z)^{s-1}}{e^z-1}dz = -2\pi i \sum_{|k|\in\mathbb{Z}\cap(0,R)} \text{Res}(f, 2\pi ik) \tag{8}$$

Hence we should evaluate the left hand side. For the small circle, we can estimate that

$$|\int_{C_\varepsilon^+} f(z)dz| \leq \int_{\pi/4}^{3\pi/2} \frac{\varepsilon^{\Re(s)-1}}{e^{\varepsilon\cos\theta}-1}d\theta \xrightarrow[\varepsilon\to 0]{} 0$$

and for the middle two lines, we take the principal branch of the logarithm Log and write

$$(-z)^{s-1} = e^{(s-1)\,\text{Log}(-z)}$$

the identity makes sense when $z \notin [0, +\infty)$, hence f is exactly well-defined on the contour, and it allows us to calculate

$$\int_{\gamma_+ \cup \gamma_-} f(z)dz = \int_\delta^T \frac{e^{(s-1)\operatorname{Log}(-x+i\delta)}}{e^{x-i\delta}-1}dx - \int_\delta^T \frac{e^{(s-1)\operatorname{Log}(-x-i\delta)}}{e^{x+i\delta}-1}dx$$

By argument principle, for any $z \in \mathbb{C} - \mathbb{R}_-$ we have

$$\operatorname{Log}|z| = \ln|z| + i \cdot \arg(z)$$

where arg is a continuous function with value in $(-\pi, \pi)$, which allows us to take limit

$$\operatorname{Log}(-x+i\delta) = \ln\sqrt{x^2+\delta^2} + i \cdot \arg(-x+i\delta) \xrightarrow[\varepsilon \to 0]{} \ln x + i\pi$$

Hence we can evaluate

$$\int_{\gamma_+ \cup \gamma_-} f(z)dz \xrightarrow[\varepsilon \to 0]{} (e^{-i\pi s} - e^{i\pi s}) \int_0^T \frac{x^{s-1}}{e^x - 1}dx$$

in particular, when R tends to infinity, we can apply the integration (7) here

$$\int_{\gamma_+ \cup \gamma_-} f(z)dz \xrightarrow[\varepsilon \to 0, R \to \infty]{} -2i \cdot \sin(\pi s)\Gamma(s)\zeta(s) \tag{9}$$

Finally we consider the large circle,

$$\begin{aligned} |\int_{C_R^-} f(z)dz| &\leq \int_{0+c}^{2\pi - c} \frac{R^{\Re(s)-1}}{e^{R\cos\theta}-1}d\theta \\ &\xrightarrow[R \to \infty, \cos(\theta) \neq 0]{} 0 \end{aligned}$$

In conclusion, by taking limits in (8), and setting (9) into it, we have

$$\begin{aligned} 2\sin(\pi s)\Gamma(s)\zeta(s) &= 2\pi \sum_{k \in \mathbb{Z} \setminus \{0\}} \operatorname{Res}(f, 2\pi i k) \\ &= 2\pi \sum_{k \in \mathbb{Z} \setminus \{0\}} (-2\pi i k)^{s-1} \\ &= 2\sin(\frac{\pi s}{2})(2\pi)^s \sum_{n \geq 1} \frac{1}{n^{1-s}} \end{aligned}$$

which is the equation given in Riemann's paper, but before the equation he had stated that zeta function is actually mermorphic in the whole complex plane with only one simple pole at $s = 1$, reason is that we just consider the integral on the path $\mathcal{H} - C_R^-$, and we can conclude

$$2\sin(\pi s)\Gamma(s)\zeta(s) = i \int_{\mathcal{H} - C_R^-} \frac{(-x)^{s-1}}{e^x - 1}dx$$

the right hand side is a entire function and we can get the analytical continuation of zeta. Therefore, Riemann's euqation can be rewritten as

$$\sin(\pi s)\Gamma(s)\zeta(s) = \sin(\frac{\pi s}{2})(2\pi)^s \zeta(1-s), \quad s \neq 0, 1 \tag{10}$$

many mordern books pointed out that the proof of Riemann is not rigorous enough, but it is not the core of the papaer, Riemann noticed the symmetry of the zeta function and gave the functional euqation

$$\pi^{-\frac{s}{2}}\Gamma(\frac{s}{2})\zeta(s) = \pi^{\frac{1-s}{2}}\Gamma(\frac{1-s}{2})\zeta(1-s)$$

Proof. The short paper do not give the proof, but with a hint of using the properties of Gamma function, so it can be seen as one of "exercises left to the reader" in this paper. Subtituting $\sin(\pi s)$ in (10) by the reflection formula (4), we have

$$\zeta(s) = 2^s\pi^{s-1}\sin(\frac{\pi s}{2})\Gamma(1-s)\zeta(1-s)$$

and subtitue $\sin(\frac{\pi s}{2})$ by the duplication formula (5), we have

$$\Gamma(\frac{s}{2})\zeta(s) = \pi^{s-\frac{1}{2}}\Gamma(\frac{1-s}{2})\zeta(1-s)$$

then it is not difficult to rearrange the equation to the desired form. □

1.2 Riemann's idea II: relation with theta function

We can found that zeta function is not so perfectly symmetric as Gamma function, the functional equation has a complicated form with other factor involved, so Riemann introduced the function we talked above, in the sense of mordern notation:

$$Z(s) = \pi^{-\frac{s}{2}}\Gamma(\frac{s}{2})\zeta(s)$$

it is so-called **completed zeta function**, with the functional equation $Z(s) = Z(1-s)$. Riemann further considered the integration representation of $Z(s)$ by some work on prevous integral (6) and (7), the idea is real stright forward: by just consider the term n^2 we have

$$\frac{\Gamma(s)}{n^{2s}} = \int_0^\infty e^{-n^2x}x^{s-1}dx$$

we expect the appearance of some factor about π, then we change variable to the right side by setting $x = \pi t$, hence

$$\frac{\Gamma(s)}{n^{2s}} = \pi^s\int_0^\infty e^{-n^2\pi t}t^{s-1}dt$$

without talking about the interchange of integration, we rearrannge and sum over n we can get

$$Z(s) = \int_0^\infty (\sum_{n\geq 1} e^{-n^2\pi t})t^{\frac{s}{2}-1}dt$$

Riemann noticed that the summation is cloesely related to the theta function, and hence he cited the formula of Jacobi to get the analyical continuation of $Z(s)$ and define the xi-function, which is the core object in his following discussion. We stoped here for the moment, the page 3 is so rich that we need to rewrite it carefully in mordern notation, the reason is as following:

- Maybe Riemann do not relaize that he actually gived the analytic continuation of zeta function again, because completed zeta function

Definition 1. The **Jacobi theta function** is defined as

$$\theta(z) = \sum_{n\in\mathbb{Z}} e^{\pi i n^2 z}$$

it is holomorphic in the upper half-plane $\mathbb{H} = \{z \in \mathbb{C} : \Im(z) > 0\}$.

It is not difficult to see that

$$|e^{\pi i n^2(x+iy)}| = |e^{-\pi n^2 y}|$$

so the series convergence abosolutely and uniformly on any compact subset $\mathbb{H}$, so θ is exactly holomoprhic, and it has a important functional equation, which was first proved by Jacobi (1829), and Riemann cited it in his paper without proof.

Lemma 2. (Jacobi) For any $z \in \mathbb{H}$, we have

$$\theta(-1/z) = \sqrt{z/i} \cdot \theta(z)$$

In particular, when $z = it$ for $t > 0$, i.e. on the positive imaginary axis, θ function has a deep relation with the completed zeta function:

Proposition 3. For $\Re(s) > 1$, completed zeta function is holomoprhic and

$$Z(s) = \frac{1}{2}\int_0^\infty (\theta(it) - 1)t^{\frac{s}{2}-1}dt$$

Proof. ζ and Γ are both holomoprhic on $\Re(s) > 1$, and $s \mapsto pi^{-\frac{s}{2}}$ is entire, so Z is holomorphic on $\Re(s) > 1$. By definition of θ function, we have

$$\frac{\theta(it) - 1}{2} = \sum_{n\geq 1} e^{-n^2\pi t}$$

which allows us to rewrite

$$\frac{1}{2}\int_0^\infty (\theta(it) - 1)t^{\frac{s}{2}-1}dt = \int_0^\infty (\sum_{n\geq 1} e^{-n^2\pi t})t^{\frac{s}{2}-1}dt$$

and if we admits the integration representation (7) and calculate as above we can get

$$Z(s) = \sum_{n\geq 1}\int_0^\infty e^{-\pi n^2 t}t^{\frac{s}{2}-1}dt$$

for $Re(s) > 1$, so we should verify the interchange of summation and integration.

$$|\sum_{n\geq 1} e^{-\pi n^2 t}t^{s/2-1}| \leq \sum_{n\geq 1} e^{-\pi n t}t^a = \frac{t^a}{e^{\pi t} - 1} \sim_{t\to\infty} t^a e^{-\pi t}$$

with $a = \Re(s)/2 - 1 > -1/2$, so the integration converges at infinity, and by Jacobi's formula we can estimate the integration near zero:

$$\theta(it) = (t)^{-1/2}\theta(i/t) = t^{-1/2}\sum_{k\in\mathbb{Z}} e^{-\pi k^2/t} \xrightarrow[t\to 0]{} t^{-1/2}$$

which allows us to get a global bound function for some constant $C > 0$:

$$|\sum_{n\geq 1} e^{-\pi n^2 t} t^{s/2-1}| \leq g(t) = \begin{cases} Ct^a e^{-\pi t} & t \geq 1 \\ Ct^{(a-1/2)} & 0 < t < 1 \end{cases}$$

and g is measurable and integrable on $(0, \infty)$, hence by Dominated Convergence Theorem we can interchange the summation and integration. $\square$

with the integral representation, completed zeta function can extend mermorphicly to the whole complex plane like what we do to Gamma function, for the sake of convience, Riemann introduced the auxillary function:

$$\psi(t) = \frac{\theta(it) - 1}{2}, \quad t > 0$$

by the Jacobi's functional equation, we can get the relfection formula

$$t^{1/2}(2\psi(t) + 1) = 2\psi(1/t) + 1, \quad t > 0 \tag{11}$$

with that we can get the following result which is roughly sketched in Riemann's paper:

Theorem 4. The completed zeta function Z has a analytic continuation to complex palne $\mathbb{C}$, and
- it only has simple poles at $s = 0$ and $s = 1$;
- residue number: $\text{Res}(Z, 0) = -1$ and $\text{Res}(Z, 1) = 1$;
- it satisfies the functional equation $Z(s) = Z(1 - s)$ for $s \neq 0, 1$.

Proof. The proof is based on the integral representation on $\Re(s) > 1$, we divide the integration into two parts and change variable

$$\begin{aligned} Z(s) &= \int_0^1 \psi(t)t^{\frac{s}{2}-1}dt + \int_1^\infty \psi(t)t^{\frac{s}{2}-1}dt \\ &= \int_1^\infty \psi(1/u)u^{-\frac{s}{2}-1}du + \int_1^\infty \psi(t)t^{\frac{s}{2}-1}dt \end{aligned}$$

then apply the formula (11) to the first term, we have

$$\begin{aligned} Z(s) &= \frac{1}{2}\int_1^\infty t^{-s/2-1/2}dt - \frac{1}{2}\int_1^\infty t^{s/2-1}dt + \int_1^\infty \psi(t)(t^{s/2-1} + t^{-s/2-1/2})dt \\ &= \frac{1}{s-1} - \frac{1}{s} + \int_1^\infty \psi(t)(t^{s/2-1} + t^{-s/2-1/2})dt \\ &= \frac{1}{s-1} - \frac{1}{s} + F(s) + F(1-s) \end{aligned}$$

where $F(s) = \int_1^\infty \psi(t)t^{s/2-1}dt$ is an entire function, which ca be proved by theorem of convergence dominated (wait to complete...). Hence we can get the analytic continuation of Z to the whole complex plane, with only simple poles at $s = 0$ and $s = 1$ left, and the residue number can be calculated by the formula above. Finally, the functional equation when $s \mapsto 1 - s$. $\square$

Corollary 5. Riemann zeta function ζ has a analytic continuation to the whole complex plane except for a simple pole at $s = 1$ with residue number 1, with functional euqation

$$\zeta(1-s) = 2(2\pi)^s \cos(\pi s/2)\Gamma(s)\zeta(s) \qquad s \neq 1, 0, -1, -2, \ldots$$

Proof. By completed zeta function, we have

$$\zeta(s) = \frac{Z(s)}{\pi^{-\frac{s}{2}}\Gamma\left(\frac{s}{2}\right)}$$

notice that Z and Γ have both simple poles at $s = 0$, so ζ is actually holomophic at $s = 0$, with only simple pole at $s = 1$ left by $Z(s)$, that because Γ has no zeros and $s \mapsto \pi^{-s/2}$ is entire. $\square$

Remark 6. The method of analytic continuation via integral representation can be generalized, it is so called Mellin transformation, an important and natural integration transform: Let f be a positive real-valued function, then its **Mellin transform** is defined as

$$\mathcal{M}(f, s) = \int_0^\infty f(t)t^{s-1}dt$$

here completd zeta function is just the Mellin transform of ψ function.

1.3 Value of Riemann's Zeta function

This section is to give a basic description of the distrubution of values of Riemann zeta function, mainly focus on the value of $\mathbb{Z}$, and then the result of Hamadard about the critial strip.

Firstly, the value at the positive odd integer of zeta function is hard and few results about that we can know. In 1978, Apery proved that $\zeta(3)$ is irrational; In 2001, Rivoal proved that there exists infinite irrational number in $\{\zeta(2k+1) \mid k \in \mathbb{N}\}$, and he gave the lower bound of the cardinal (as the scale of $\log k$); Another folklore conjecture about the irrationality is that $\zeta(2k+1)$ is algebraical independent over $\mathbb{Q}$, it is the corollary of the famous **Grothendieck's period conjecture**, which is still open.

The simplest case is at the negative even integer, which is concluded in Riemann's paper as a samll comments: $\zeta(-2k) = 0$ for any $k \in \mathbb{N}$.

Proof. The proof is based on the functional equation of zeta function, it refers to corollary 5, just take $1 - s = -2k$, for $k \in \mathbb{N}$; then we can find that $\cos(\pi s/2) = 0$, and we can conclude the result. A little remark here is that the method can not be extended to the positive odd integer, the reason is that Γ is only well-defined on the positive integer, which shows that the lack of information is the diffculty of the question at positive odd integer. $\square$

The value at negative odd intgers and at positive even integers are the same story, that is clear by the functional equation, if we set $1 - s = -2k - 1$ for $k \in \mathbb{N}$, then we can conclude

$$\zeta(-2k-1) = 2(2\pi)^{2k+4}(-1)^{k+1}(2k+1)!\zeta(2k+2)$$

so our attention will focus on the values of positive even integers firstly. Indeed the values of zeta function at $\mathbb{Z}$ has a deep relation with Bernoulli's number, formally we define:

Definition 7. The function $z \mapsto \frac{z}{e^z-1}$ is holomorphic at the voisinage of zero, with the expansion of the series:

$$\frac{z}{e^z - 1} = \sum_{n \geq 0} \frac{B_n}{n!} z^n$$

the n-th **Bernoulli's number** is defined as B_n, with respect to the coefficient of the series.

The original definition of Bernoulli's number is given by Jacob Bernoulli in his book "Ars Conjectandi" (1713), he defined the number by the summation of powers:

$$\sum_{k=1}^{m} k^n = \frac{1}{n+1} \sum_{j=0}^{n} \binom{n+1}{j} B_j \cdot m^{n+1-j}$$

for any $n, m \in \mathbb{N}$, and the two definitions can be shown to be equivalent by the method of **generating function**. The definition of power series is easy for us to calculate the values of Bernoulli's number, for Example

$$B_0 = 1, \quad B_1 = \frac{-1}{2}, \quad B_2 = \frac{1}{6}, \quad B_3 = 0, \quad B_4 = \frac{-1}{30} \quad \ldots$$

Lemma 8. The Bernoulli's number is rational, and $B_{2k+3} = 0$ for any $k \in \mathbb{N}$.

Proof. Obeserve that

$$\frac{ze^z}{e^z - 1} = \frac{-z}{e^{-z} - 1}$$

which allows us to write

$$z = \frac{ze^z}{e^z - 1} - \frac{z}{e^z - 1} = \sum_{n \geq 0} ((-1)^n - 1) \frac{B_n}{n!} \cdot z^n$$

at the voisinage of zero, hence by comparing the coefficients we can conclude that $B_0 = 1$ and $B_n = 0$ for any odd integer $n \geq 3$. To prove the rationalty, by the properties of analytical function we know

$$B_n = n! \frac{d^n}{dz^n} \left(\frac{z}{e^z - 1} \right) \bigg|_{z=0}$$

the derivatives can be proved to be a rational number by induction on n, hence the result holds true. ∎

With the notation of the Bernoulli's number, the values can be written as a neat form

Proposition 9. Let $k \in \mathbb{N}$, then

$$\zeta(2k) = (-1)^{k+1} \frac{(2\pi)^{2k}}{2(2k)!} \cdot B_{2k}$$

$$\zeta(-2k+1) = \frac{B_{2k}}{-2k}$$

which shows that the values at negative odd integers are rational numbers, while the values at positive even integers are transcendental numbers (rational number $\times$ the factor π).

Proof. One way to prove the result is to use the formula of cotangent function

$$\pi z \cot(\pi z) = 1 + \sum_{n=1}^{\infty} \frac{2z^2}{z^2 - n^2} \qquad z \in \mathbb{C} - \mathbb{Z}$$

using gemetric series we can rewrite the formula

$$\begin{aligned}\pi z \cot(\pi z) &= 1 + 2z^2 \sum_{n\geq 1} \frac{-1}{n^2} \cdot \frac{1}{1 - z^2/n^2} \\ &= 1 - 2z^2 \sum_{n\geq 1} \frac{1}{n^2} \sum_{k\geq 0} \left(\frac{z^2}{n^2}\right)^k \\ &= 1 - 2\sum_{k\geq 1} \zeta(2k) z^{2k}\end{aligned}$$

here the formula holds true for $|z| < 1$, and the last equality holds by Fubini's theorem, so we get a power series expansion at zeros. To finish the proof, we write the trigonometric function in terms of exponential function, then we can calculate

$$z\cot(z) = \frac{2iz}{e^{2iz} - 1} + iz = 1 + \sum_{k=1}^{\infty} (-1)^k \frac{2^{2k}}{(2k)!} B_{2k} \cdot z^{2k}$$

which allows us to get another power series expansion at zeros, if we replace z by πz and compare the two expansions, then we can conclude the first formula. The second formula can be concluded by the functional equation of zeta function. □

2 L-function

2.1 Some preparation

2.2 Functional equation of L-function

Review the functional equation of Riemann zeta function and its analytical continuation, we can conclude the method by two steps: Firstly, we find an integral representation of zeta function by the Gamma function, and rerrange the integration to get the integral reprsentation of completed zeta function, it gives a relation with theta function; Secondly, we use the functional equation of theta function to rewrite the integral, which allows us to get the functional equation of completed zeta function, and hence the functional equation of zeta function.

For any Dirichlet character χ modulo q, there exists a conductor q' which divides q such that

$$\chi = \chi' \cdot 1$$

with χ' a primitive Dirichlet character modulo q', and 1 the trivial character modulo q. Hence for any L-function on χ, we can write

$$L(s, \chi) = L(s, \chi') \cdot \prod_{p|q} (1 - \chi(p) p^{-s})$$

the last term defines a entire function, so the analytic continuation will be reduced to the case of primitive character.

The method is so natural that it can be generalized to the L-function, and we start from the integral

$$\pi^{-\frac{s}{2}}\Gamma(\frac{s}{2})\frac{1}{n^s} = \int_0^\infty e^{-\pi n^2 t} t^{\frac{s}{2}-1} dt \qquad \Re(s) > 1$$

if we sum over n, that will be what we do in the last section, but now we want to sum over n with **some primitive Dirichlet character** χ modulo q, and we expect the auxillary function is a certain arithemetic function, with at least the information of the character (periodic q).

$$(q/\pi)^{s/2}\Gamma(\frac{s}{2})\frac{\chi(n)}{n^s} = \frac{1}{\sqrt{q}}\int_0^\infty [\chi(n)e^{-\pi n^2 t/p}] t^{\frac{s}{2}-1} dt \tag{12}$$

what we do is to change the variable by setting $t \to t/q$, hence we can sum over n and get

$$(q/\pi)^{s/2}\Gamma(\frac{s}{2})L(s,\chi) = \frac{1}{\sqrt{q}}\int_0^\infty [\sum_{n\geq 1}\chi(n)e^{-\pi n^2 t/q}] t^{\frac{s}{2}-1} dt$$

we can extend the sum to the whole integer, but it depends on the values of $\chi(-1)$. If $\chi(-1) = 1$, then we can write

$$\sum_{n\geq 1}\chi(n)e^{-\pi n^2 t/q} = \frac{1}{2}\sum_{n\in\mathbb{Z}}\chi(n)e^{-\pi n^2 t/q}$$

and if $\chi(-1) = -1$ we find that the right term above is zero, it makes no sense, hence we should modify the integration (12) a little to avoid the problem, we can introduce a translation on s by setting $s \to s+1$, then we can write

$$(q/\pi)^{(s+1)/2}\Gamma(\frac{s+1}{2})\frac{\chi(n)}{n^s} = \frac{1}{\sqrt{q}}\int_0^\infty [n\chi(n)e^{-\pi n^2 t/p}] t^{\frac{s-1}{2}} dt \tag{13}$$

then the sum over n allows us to get

$$\sum_{n\geq 1} n\chi(n)e^{-\pi n^2 t/q} = \frac{1}{2}\sum_{n\in\mathbb{Z}} n\chi(n)e^{-\pi n^2 t/q}$$

It is the basic idea of the proof, similarly we introduce the **auxillary function**

$$\psi(t,\chi) = \sum_{n\in\mathbb{Z}} n^{\mathfrak{v}}\chi(n)e^{-\pi n^2 t/q}$$

with $\mathfrak{v} = \frac{1-\chi(-1)}{2}$, then we can deduce the functional equation via Poisson's formula.

Lemma 10. let $t > 0$, then ψ satisfies the equation

$$\tau(\bar{\chi})\psi(t,\chi) = (i/t)^{\mathfrak{v}}\sqrt{q/t}\cdot\psi(\bar{\chi}, 1/t)$$

where $\tau(\chi) = G(1,\chi)$ as the Gauss sum of χ.

Proof. We consider the $f(x) = e^{-\pi x^2 t}$ for some $t > 0$, and then we can calculate

$$\hat{f}(s) = \frac{1}{\sqrt{t}} f(\frac{s}{t}) = \frac{1}{\sqrt{t}} e^{-\pi s^2/t}$$

hence by poisson's formula we have

$$\sum_{n\in\mathbb{Z}} e^{-\pi(n+a)^2 t} = \frac{1}{\sqrt{t}} \sum_{n\in\mathbb{Z}} e^{-\pi n^2/t} e^{2\pi i n a} \tag{14}$$

now we consider χ is a arithemetic function on $\mathbb{Z}$ with period q, then we can write

$$\chi(n) = \frac{\tau(\chi)}{q} \sum_{m=1}^{q} \bar{\chi}(m) e^{2\pi i m n/q}$$

by the primitive character $\overline{\tau(\chi)} = \chi(-1)\tau(\bar{\chi})$, we have

$$\tau(\bar{\chi})\chi(n) = \sum_{m=1}^{q} \bar{\chi}(m) e^{2\pi i m n/q}$$

hence we can calculate

$$\begin{aligned}
\tau(\bar{\chi})\psi(t,\chi) &= \sum_{n\in\mathbb{Z}} n^{\mathfrak{v}} \tau(\bar{\chi})\chi(n) e^{-\pi n^2 t/q} \\
&= \sum_{m=1}^{q} \bar{\chi}(m) \sum_{n\in\mathbb{Z}} n^{\mathfrak{v}} e^{-\pi n^2 t/q} e^{2\pi i n m/q}
\end{aligned}$$

when $\chi(-1) = 1$, i.e. $\mathfrak{v} = 0$, we can apply the Poisson's formula (14) to get

$$\begin{aligned}
\tau(\bar{\chi})\psi(t,\chi) &= \sqrt{q/t} \cdot \sum_{m=1}^{q} \bar{\chi}(m) \cdot \sum_{n\in\mathbb{Z}} e^{-\pi(qn+m)^2/qt} \\
&= \sqrt{q/t} \cdot \sum_{l\in\mathbb{Z}} \bar{\chi}(l) e^{-\pi l^2/t} \\
&= \sqrt{q/t} \cdot \psi(\bar{\chi}, 1/t)
\end{aligned}$$

when $\chi(-1) = -1$, i.e. $\mathfrak{v} = 1$, we can modify the poisson's formula (14) by taking derivative of $f(x)$, then we have

$$it^{3/2} \sum_{n\in\mathbb{Z}} (n+a) e^{-\pi(n+a)^2 t} = \sum_{n\in\mathbb{Z}} n e^{-\pi n^2/t} e^{2\pi i n a}$$

then apply it we can get

$$\begin{aligned}
\tau(\bar{\chi})\psi(t,\chi) &= \sum_{m=1}^{q} \bar{\chi}(m) \cdot i(q/t)^{3/2} \sum_{n\in\mathbb{Z}} (n + \frac{m}{q}) e^{-\pi(qn+m)^2/qt} \\
&= (i/t)\sqrt{q/t} \cdot \sum_{l\in\mathbb{Z}} l \cdot \bar{\chi}(l) e^{-\pi l^2/t} \\
&= (i/t)\sqrt{q/t} \cdot \psi(\bar{\chi}, 1/t)
\end{aligned}$$

□

With the functional equation of ψ, it is sufficient to get the functional equation of L function, firstly we define the **completed L function** by

$$\Lambda(s,\chi) = (q/\pi)^{\frac{s+\mathfrak{v}}{2}}\Gamma(\frac{s+\mathfrak{v}}{2})L(s,\chi)$$

by (12) and (13), we can write down the integral representation

$$\Lambda(s,\chi) = \frac{1}{2\sqrt{q}}\int_0^\infty \psi(t,\chi)t^{\frac{s+\mathfrak{v}}{2}-1}dt$$

when $\Re(s) > 1$, and then by the functional equation of ψ, we can get

Theorem 11. For any **non-principal** primitive Dirichlet character χ modulo q, the completed L function $\Lambda(s,\chi)$ has an analytic continuation to the whole complex plane, and it satisfies the functional equation

$$\Lambda(1-s,\bar{\chi}) = \frac{i^{\mathfrak{v}}\sqrt{q}}{\tau(\chi)}\Lambda(s,\chi)$$

Proof. when $\chi(-1) = 1$, i.e. $\mathfrak{v} = 0$, we have

$$\begin{aligned}
2\sqrt{q}\Lambda(s,\chi) &= \int_0^1 \psi(t,\chi)t^{\frac{s}{2}-1}dt + \int_1^\infty \psi(t,\chi)t^{\frac{s}{2}-1}dt \\
&= \int_1^\infty \psi(1/u,\chi)u^{-\frac{s}{2}-1}du + \int_1^\infty \psi(t,\chi)t^{\frac{s}{2}-1}dt \\
&= \frac{\tau(\chi)}{\sqrt{q}}\int_1^\infty \psi(u,\bar{\chi})u^{\frac{-s-1}{2}}du + \int_1^\infty \psi(t,\chi)t^{\frac{s}{2}-1}dt
\end{aligned}$$

notice that in this case we have $\overline{\tau(\chi)} = \tau(\bar{\chi})$, so we can write

$$\begin{aligned}
2\sqrt{q}\Lambda(s,\chi) &= \frac{\tau(\chi)}{\sqrt{q}}[\int_1^\infty \psi(u,\bar{\chi})u^{\frac{-s-1}{2}}du + \frac{\tau(\bar{\chi})}{\sqrt{q}}\int_1^\infty \psi(t,\bar{\chi})t^{\frac{-s-1}{2}-1}dt] \\
&= \frac{\tau(\chi)}{\sqrt{q}} \cdot 2\sqrt{q}\Lambda(1-s,\bar{\chi})
\end{aligned}$$

Finally, when $\chi(-1) = -1$, i.e. $\mathfrak{v} = 1$, we have

$$\begin{aligned}
2\sqrt{q}\Lambda(s,\chi) &= \int_0^1 \psi(t,\chi)t^{\frac{s-1}{2}}dt + \int_1^\infty \psi(t,\chi)t^{\frac{s-1}{2}}dt \\
&= \int_1^\infty \psi(1/u,\chi)u^{-\frac{s+3}{2}}du + \int_1^\infty \psi(t,\chi)t^{\frac{s-1}{2}}dt \\
&= \frac{\tau(\chi)}{i\sqrt{q}}\int_1^\infty \psi(u,\bar{\chi})u^{-\frac{s}{2}}du + \int_1^\infty \psi(t,\chi)t^{\frac{s-1}{2}}dt
\end{aligned}$$

and notice that in this case we have $\overline{\tau(\chi)} = -\tau(\bar{\chi})$, so we can write

$$\begin{aligned}
2\sqrt{q}\Lambda(s,\chi) &= \frac{\tau(\chi)}{i\sqrt{q}}[\int_1^\infty \psi(u,\bar{\chi})u^{-\frac{s}{2}}du + \frac{\tau(\bar{\chi})}{i\sqrt{q}}\int_1^\infty \psi(t,\bar{\chi})t^{\frac{s-1}{2}}dt] \\
&= \frac{\tau(\chi)}{i\sqrt{q}} \cdot 2\sqrt{q}\Lambda(1-s,\bar{\chi})
\end{aligned}$$

□

similarly, we can conclude that the

3 Dedekind zeta function

3.1 Lattice

Definition 12. Let $(V, \langle \cdot, \cdot \rangle)$ be a finite-dimensional inner product space over $\mathbb{R}$:

- A **lattice** L in V is a discrete subgroup of V.

- A lattice L is called **complete** if it is a free $\mathbb{Z}$-module of rank $n = \dim V$ with basis $\{v_1, \ldots, v_n\}$, its fundamental domain in defined as

$$P = \{\sum_i^n c_i v_i \mid c_i \in [0, 1)\}$$

- For any lattice $L \subset V$, the dual lattice under the inner product is defined as

$$L^* = \{v \in V \mid \langle v, w \rangle \in \mathbb{Z}, \forall w \in L\}$$

Notice that V/L is compact if and only if L is complete, hence we can define the **volume** of V/L by

$$\mathrm{Vol}(V/L) = m(P) = \sqrt{\det G} \cdot l(P)$$

where m is unique Haar measure induced by the inner product while l is the usuel lebesgue measure, and G is the Gram matrix with entries $\langle v_i, v_j \rangle$.

Lemma 13.

$$\mathrm{Vol}(V/L) \cdot \mathrm{Vol}(V/L^*) = 1$$

Proof. □

Theorem 14 (Poisson's formula for lattice)**.** Let L be a complete lattice in V, then for any Schwartz function f on V, we have

$$\sum_{v \in L} f(v) = \frac{1}{\mathrm{vol}(V/L)} \sum_{v^* \in L^*} \hat{f}(v^*)$$

Proof. □

Corollary 15. Let A be a positive-definite symmetric matrix, and $L \subset V$ be a complete lattice, then

$$\sum_{v \in L} e^{-\pi \cdot v^t A v} = \frac{\det A^{-1/2}}{\mathrm{vol}(V/L)} \sum_{v \in L^*} e^{-\pi \cdot v^t A^{-1} v}$$

In particular, if $A = \mathrm{dig}(c_1, \ldots, c_n)$ and $L = \mathbb{Z}^n$, then

$$\sum_{v \in \mathbb{Z}^n} e^{-\pi \sum_i c_i v_i^2} = \frac{1}{\sqrt{c_1 \cdots c_n}} \sum_{v \in \mathbb{Z}^n} e^{-\pi \sum_i v_i^2 / c_i}$$

Proof. $\square$

Now we consider the case of number field, suppose that $k/\mathbb{Q}$ is a number field with degree N, and we embded k into $\mathbb{R}^N$ by

$$\sigma : k \to \mathbb{R}^s \times \mathbb{C}^t, \quad x \mapsto (\sigma_i(x))_i$$

view $\mathbb{C}$ as a real vector space and then we can define inner product by trace

$$\langle \sigma(x), \sigma(y) \rangle = \mathrm{Tr}_{k/\mathbb{Q}}(xy)$$

Notice that $\mathcal{O}_k$ is a free $\mathbb{Z}$-module of rank N, so $\sigma(\mathcal{O}_k)$ is a complete lattice, and it motivates us to define its dual lattice.

Definition 16. The inverse different of k is a fractional ideal defined by

$$\mathfrak{d}^{-1} = \{x \in k \mid \mathrm{Tr}_{k/\mathbb{Q}}(x\mathcal{O}_k) \subset \mathbb{Z}\}$$

Lemma 17. Along the notion above,

1. $\sigma(\mathfrak{d}^{-1})$ is the dual lattice of $\sigma(\mathcal{O}_k)$ under the inner product defined by trace.
2. $\mathfrak{d} := (\mathfrak{d}^{-1})^{-1}$ is an ideal of $\mathcal{O}_k$, and it is called the **different** of k.
3. The norm of different is the discriminant of the field $N(\mathfrak{d}) = |d_k|$
4. For any ideal $\mathfrak{a}$, we set $\mathfrak{a}^\vee = \mathfrak{d}^{-1}\mathfrak{a}^{-1}$, then $\sigma(a^\vee)$ is the dual lattice of $\sigma(\mathfrak{a})$.

Proposition 18. Let $\mathfrak{a}$ be an fractional ideal of k, and define

$$\Theta(c, \mathfrak{a}) = \sum_{z \in \mathfrak{a}} \exp(-\pi \sum_{c_i})$$

where $\|\cdot\|$ is the norm defined by trace paring, then we have the equation

$$\Theta(z, \mathfrak{a}) = \frac{z^{-N/2}}{N(\mathfrak{a})\sqrt{|d_k|}} \Theta(1/z, \mathfrak{a}^\vee)$$

3.2 Functional Equation

Suppose that $k/\mathbb{Q}$ is a number field with degree N, we make the convention for the notion:

1. r and t are the number of real and complex embedding of k respectively, so $N = r + 2t$;
2. $C_k = J_k/P_k$ is the ideal class group of k, and $h_k = |C_k|$ is the class number of k;
3. w_k is the number of roots of unity in k, r_k is the regulator of k, and d_k is the discriminant of k.

For any ideal class $R \in C_k$, we take an representative $\mathfrak{a}$ of R^{-1}, then for any integral ideal $\mathfrak{b} \in R$ we have a well-defined applicaion to P_k:

$$\mathfrak{b} \mapsto \mathfrak{a}\mathfrak{b} = (x)$$

notice that $\mathfrak{b} \subset \mathcal{O}_k$ as an integral ideal, so $x \in \mathfrak{a}^*$. Conversely, for any $x \in \mathfrak{a}^*$ we can set $\mathfrak{b} = a^{-1}(x)$, then $\mathfrak{b} \subset \mathcal{O}_k^*$ implies $\mathfrak{b}$ is exactly a integral ideal, so the application is surjective to $\mathfrak{a}^*$. For a bijection, we consider an equivalence relation on $\mathfrak{a}^*$ by setting $x \sim x'$ if and only if $x' = ux$ for some unit u, i.e. two elements are associative, it is equivalent to the the action of $\mathcal{O}_k^*$ on $\mathfrak{a}^*$, hence we can get a bijection

$$\{\text{The integral ideal of } R\} \longleftrightarrow \mathfrak{a}^*/\mathcal{O}_k^*$$

the right term is the orbit space of the action. Hence we can rewrite the Dedekind zeta function

$$\zeta_k(s) = \sum_{R \in C_k} \zeta(s, R)$$

as the sum of function on ideal classes.

$$\zeta(s, R) = N(\mathfrak{a})^s \sum_{x \in \mathfrak{a}^*/\mathcal{O}_k^*} \frac{1}{|N(x)|^s} = (d_{\mathfrak{a}}/d_k)^{1/2} \sum \dots$$

where $\mathfrak{a}$ is the representative of R^{-1}, Hence it motivates us to find the functional equation of zeta function on ideal class.

We consider the minkowski space $\mathbb{R}^r \times \mathbb{C}^t$, then in it we can denote the norm as following

$$|N(x)| = \prod_v |x_v|^{a_v}$$

with $x_v = \sigma_v(x)$ and v runs over embeddings modulo conjugation, and

$$a_v = \begin{cases} 1 & \text{if } \sigma_v \text{ is a real embedding} \\ 2 & \text{if } \sigma_v \text{ is a complex embedding} \end{cases}$$

Proposition 19. With the notion above, we set

$$A = 2^{-t} d_k^{-1/2} \pi^{-N/2}$$

and let

$$\zeta(s, \infty) = A^s \Gamma(s/2)^r \Gamma(s)^t$$

then define the completed ideal class zeta function

$$\Lambda(s, R) = \zeta(s, \infty)\zeta(s, R)$$

Then $\Lambda(s, R)$ attains an analytical continuation on complex plane with only simple poles at $s = 0$ and $s = 1$, and it satisfies a functional equation for any $s \neq 0, 1$

$$\Lambda(1 - s, R^{-1}) = \Lambda(s, R)$$

Proof. (1) Representation of function: We consider the haar measure on $\mathbb{R}_{>}0$ and definition of Gamma function, it sufficient to prove that

$$\frac{\Gamma(s/2)}{a^s} = \int_{\mathbb{R}_{>0}} e^{-a^2 y} y^{s/2} \frac{dy}{y}$$

Applying this formula to $a^2 = ...$, it is enough to get

$$(\pi^{-1/2} d_k^{1/2N} N(\mathfrak{a})^{1/N})^s = \int_{\mathbb{R}_{>0}} \exp(-\pi d_{\mathfrak{a}}^{-1/N} |x_v|^2 y) y^{s/2} \frac{dy}{y} \tag{15}$$

For real parts we will multiply (15) r times, while for complex embedding, we should let $s \to 2s$, and change variable in integral such that $y \mapsto 2y$. The first operation is because we will only multiply complex parts t times while $N = r + 2t$; The second operation is necessary, because we expect a fourier transform with respects to the inner product in Minkowski space, hence for complex parts $|x_v|$ we have

$$(2^{-1} \pi^{-1} d_k^{1/N} N(\mathfrak{a})^{2/N})^s = \int_{\mathbb{R}_{>0}} \exp(-\pi d_{\mathfrak{a}}^{-1/N} \cdot 2|x_v|^2 y) y^s \frac{dy}{y} \tag{16}$$

As we mentioned, consider $(15) \times r + (16) \times t$ and sum over $\mathfrak{a}^*/\mathcal{O}_k^*$ we have

$$\Lambda(s, R) = \int_G \sum_x \exp(-\pi d_{\mathfrak{a}}^{1/N} \cdot \langle x, y \rangle_m) \|y\|^{s/2} d^* y$$

where $G = \mathbb{R}_{>0}^{s+t}$ as a locally compact group under the restricted product with Harr measure such that

$$d^* y = \prod_v \frac{dy_v}{y_v}$$

(2) Decomposition of integration: We now consider the spherical decomposition of group $G = \mathbb{R}_{>0} \times G_0$ with

$$G_0 = \{x \in G \mid \|x\| = 1\}$$

i.e. we set $y = l^{1/N} c$ with $l = \|y\|$, then we have

$$\Lambda(s, R) = \int_{\mathbb{R}>0} \int_{G_0} \exp(-\pi d_{\mathfrak{a}}^{1/N} \cdot \langle x, l^{1/N} c \rangle) d^* c \quad t^{s/2} \frac{dt}{t}$$

□

4 Hecke character

4.1 Größencharakter

As we have done for Riemann zeta function, we hope to consider dedekind zeta function with character, so it is natural to consider the character

$$\chi_{\mathfrak{m}} : (\mathcal{O}_k/\mathfrak{m})^* \to \mathbb{C}^*$$

where $\mathfrak{m}$ is an ideal in $\mathcal{O}_k$. Notice that $\mathcal{O}_k/\mathfrak{m}$ is finite, so the character is finite onto $\mathbb{S}^1$ exactly. However, ring of integers $\mathcal{O}_k$ is not always a UFD, so it is natural to consider the character on ideals.

Firstly set some notation here:

- $J(\mathfrak{m}) = \{\mathfrak{a} \in J_k \mid v_{\mathfrak{p}}(\mathfrak{a}) = 0,\ \forall \mathfrak{p} \mid \mathfrak{m}\}$ fractional ideals coprime with $\mathfrak{m}$

- $P(\mathfrak{m}) = J(\mathfrak{m}) \cap P_k$ principal ideals coprime with $\mathfrak{m}$

- $\iota : k^* \to P_k, x \mapsto (x)$ is a surjective group homomorphism with kernel $\mathcal{O}_k^*$ the group of units.

- $k(\mathfrak{m}) = \iota^{-1}(P(\mathfrak{m}))$ the pre-image, it is a subgroup of k^*

The first things is to translate the additive group structure to multiplicative group structure, from elements to ideals

To determine which elements are inversible in $\mathcal{O}_k/\mathfrak{m}$, we can write

$$\begin{aligned} \bar{a} \in (\mathcal{O}_k/\mathfrak{m})^* &\iff \exists b \in \mathcal{O}_k, ab = 1 \mod \mathfrak{m} \\ &\implies 1 = ab + (ab-1) \in (a) + \mathfrak{m} \\ &\implies (a) + \mathfrak{m} = \mathcal{O}_k \end{aligned}$$

conversely, if $(a) + \mathfrak{m} = \mathcal{O}_k$, then there exists $x \in \mathfrak{m}$ and $b \in \mathcal{O}_k$ such that $1 = ab + x$, so $ab - 1 = x = 1 \mod \mathfrak{m}$, hence we have

$$\mathcal{O}_k(\mathfrak{m}) = \{a \in \mathcal{O}_k \mid (a) + \mathfrak{m} = \mathcal{O}_k\} = k(\mathfrak{m}) \cap \mathcal{O}_k$$

where $\mathcal{O}_k(\mathfrak{m})$ is exactly the set of elements which are invertible in $\mathcal{O}_k/\mathfrak{m}$, so it restricts us to consider characters on $k(\mathfrak{m})$.

Then we rewrite the residue class, for any $a, b \in k(\mathfrak{m})$

$$\begin{aligned} a = b \mod \mathfrak{m} &\iff a - b \in \mathfrak{m} \\ &\iff \mathfrak{m} | (a-b) \\ &\iff v_{\mathfrak{p}}(a-b) \geq v_{\mathfrak{p}}(\mathfrak{m}), \forall \mathfrak{p}(|\mathfrak{m}) \\ &\iff a - b \in S \end{aligned}$$

where $S = \{x \in k(\mathfrak{m}) \mid v_{\mathfrak{p}}(x) \geq v_{\mathfrak{p}}(\mathfrak{m}), \forall \mathfrak{p}(|\mathfrak{m})\}$ as the set of elements $x = 0 \mod \mathfrak{m}$. We can investigate the structure of S to find that

- $a, b \in S$ then $ab \in S$: $v_{\mathfrak{p}}(ab) = v_{\mathfrak{p}}(a) + v_{\mathfrak{p}}(b) \geq 2v_{\mathfrak{p}}(\mathfrak{m}) \geq v_{\mathfrak{p}}(\mathfrak{m})$.
- S is not a subgroup since for any $a \in S$, $v_{\mathfrak{p}}(a^{-1}) = -v_{\mathfrak{p}}(a) \leq 0$.
- $aS = S$ for any $a \in k(\mathfrak{m})$: for any $s \in S$, we have

$$v_{\mathfrak{p}}(as) = v_{\mathfrak{p}}(a) + v_{\mathfrak{p}}(s) \geq 0 + v_{\mathfrak{p}}(\mathfrak{m})$$

where $\mathfrak{p}|\mathfrak{m}$, so it is enough to get $as \in S$. Conversely, $a^{-1} \in k(\mathfrak{m})$ since $k(\mathfrak{m})$ is a group, hence for any $s \in S$,

$$v_{\mathfrak{p}}(s) = v_{\mathfrak{p}}(a^{-1}) + v_{\mathfrak{p}}(as) = v_{\mathfrak{p}}(as)$$

which implies $as \in S$.

Hence S is not a good choice to rewrite the residue class, but by the final point above, we can keep going on

$$\begin{aligned} a = b \mod \mathfrak{m} &\iff a \in b + S = b + bS = b(1+S) \\ &\iff a/b \in 1 + S \end{aligned}$$

so here the relation given by addition turns to a relation given by multiplication, and hence we can define

$$\begin{aligned} k_{\mathfrak{m}} = 1 + S &= \{x \in k(\mathfrak{m}) \mid v_{\mathfrak{p}}(x-1) \geq v_{\mathfrak{p}}(\mathfrak{m}), \forall \mathfrak{p}(|\mathfrak{m})\} \\ &= \{x \in k(\mathfrak{m}) \mid x = 1 \mod \mathfrak{m}\} \end{aligned}$$

and then we can conclude the following things:

Lemma 20. $k_{\mathfrak{m}}$ is a subgroup of $k(\mathfrak{m})$, and there exists an isomorphism

$$(\mathcal{O}_k/\mathfrak{m})^* \to k(\mathfrak{m})/k_{\mathfrak{m}}, \quad x + \mathfrak{m} \mapsto x \cdot k_{\mathfrak{m}}$$

hence any character $\chi_{\mathfrak{m}}$ on $(\mathcal{O}_k/\mathfrak{m})^*$ can be lifted to a character f on $k(\mathfrak{m})$ with $f|_{k_{\mathfrak{m}}} = 1$; Conversely, any such character can be pushed down to a character on $(\mathcal{O}_k/\mathfrak{m})^*$.

Now we consider the principal ideals by natural map:

$$\iota : k(\mathfrak{m}) \to P(\mathfrak{m}), \quad x \mapsto (x) = x\mathcal{O}_k$$

then the kernel is $\mathcal{O}_k^* \cap k(\mathfrak{m}) = \mathcal{O}_k^*$ such that we can consider the following diagram

$$\begin{tikzcd} k(\mathfrak{m}) \arrow[r, "\iota"] \arrow[d] & P(\mathfrak{m}) \arrow[d, "\pi"] \\ k(\mathfrak{m})/\mathcal{O}_k^* \cdot k_{\mathfrak{m}} \arrow[r, dashed, "\sim"] & P(\mathfrak{m})/P_{\mathfrak{m}} \end{tikzcd}$$

here the kernel is calculated as following

$$\begin{aligned} \ker(\pi \circ \iota) &= \{x \in k(\mathfrak{m}) \mid (x) \in P_{\mathfrak{m}}\} \\ &= \{x \in k(\mathfrak{m}) \mid \exists u \in \mathcal{O}_k^*, ux \in k_{\mathfrak{m}}\} \\ &= \mathcal{O}_k^* \cdot k_{\mathfrak{m}} \end{aligned}$$

Then we consider the infinite places, it is necessary to handle unit here.

We set the minkowski's space $k_\infty^* = (\mathbb{R}^*)^r \times (\mathbb{C}^*)^s$, and then

$$k^* \xhookrightarrow{i} k_\infty^* \xrightarrow{\chi_\infty} \mathbb{C}^*$$

we consider the continous character here, its form can be determined as following

Lemma 21. Along the notation

1. Any continous character $\mathbb{R}^* \to \mathbb{C}^*$ is of the form
$$\chi(t) = \chi(\mathrm{sign}(t)) \cdot |t|^\lambda$$
with $\lambda \in \mathbb{C}$ and $\mathrm{sign}(t) = \frac{t}{|t|} = \pm 1$.

2. Any continous character $\mathbb{C}^* \to \mathbb{C}^*$ is of the form
$$\chi(z) = (\frac{z}{|z|})^n \cdot |z|^\lambda$$
with $\lambda \in \mathbb{C}$ and $n \in \mathbb{Z}$.

3. Any continous character $\chi_\infty : k_\infty^* \to \mathbb{C}^*$ is of the form
$$\chi_\infty(x) = w(x) \cdot \prod_v |x_v|^{it_v}$$
where $t_v \in \mathbb{R}$ while w is a finite character on the group of units $U_\infty = \{x \in k_\infty^* \mid |x_v| = 1, \forall v\}$.

Proof. □

The reason why we consider the character on infinite places is as following: for a character $\chi_\mathfrak{m} : (\mathcal{O}_k/\mathfrak{m})^* \to \mathbb{S}^1$, we consider its induced character $f : k(\mathfrak{m}) \to \mathbb{S}^1$ with $f|_{k_\mathfrak{m}} = 1$. It motivates us to define as following
$$\chi : P(\mathfrak{m}) \to \mathbb{S}^1, \quad (a) \mapsto f(a)$$
however, here are some problems:

- Any $\mathfrak{a} \in J(\mathfrak{m})$, then $\mathfrak{a}^h = (a) \in P(\mathfrak{m})$ with h here the class number, so we can try to extend χ on $P(\mathfrak{m})$ by writing
$$\chi(\mathfrak{a})^h = \chi((a)) = f(a)$$
but the extension is not unique since we can choose any h-th root of $f(a)$ to be $\chi(\mathfrak{a})$.

- Above map we defined is not well-defined: if $(a) = (b) = (ua)$ for some unit u, then we have
$$\chi((a)) = \chi((ua)) \iff f(u) = 1$$
but f can be non-trival on units, so problems comes out.

- To solve the problem, we try to invite the character on infinite places satisfying
$$\chi((a)) = f(a) \cdot \chi_\infty(i(a)) \quad \forall a \in k(\mathfrak{m})$$
in this case if $u \in \mathcal{O}_k^*$
$$1 = f(u) \cdot \chi_\infty(i(u))$$
and if $a \in k_\mathfrak{m}$ such that $f(a) = 1$
$$\chi((a)) = \chi_\infty(i(a))$$

this gives the restrictions on χ, or a commute digram as following

$$\begin{tikzcd} k_{\mathfrak{m}} \arrow[r, "\iota"] \arrow[d, hook] & P(\mathfrak{m}) \arrow[d, "\chi"] \\ k_\infty^* \arrow[r, "\chi_\infty"'] & \mathbb{S}^1 \end{tikzcd}$$

Theorem 22 (Weak apporximation theorem)**.**
Let $v_1, \dots, v_m$ be pairwise inequivalent **absolute values** of the field k, and $a_1, \dots, a_m \in k$ be any elements, then for any $\epsilon > 0$, there exists an element $x \in k$ such that

$$v_i(x - a_i) < \epsilon \quad \text{for} i = 1, \dots, m$$

Proof. notice that for any i, there exists an element y_i such that

$$v_j(y_i) \begin{cases} > 1 & j = i \\ < 1 & j \neq i \end{cases}$$

hence we can consider the sequence $x_n^{(i)} = \frac{y_i^n}{1+y_i^n}$, then

$$v_j(x_n^{(i)}) \xrightarrow{n\to\infty} \begin{cases} 1 & j = i \\ 0 & j \neq i \end{cases}$$

Hence we can consider $x_n = \sum_{k=1}^m a_k x_n^{(k)}$, then for any i we have

$$\begin{aligned} v_i(x_n - a_i) &= v_i(a_i) v_i(1 - x_n^{(i)}) + \sum_{k \neq i} v_i(a_k) v_i(x_n^{(k)}) \\ &\xrightarrow{n\to\infty} v_i(a_i) \cdot 0 + 0 = 0 \end{aligned}$$

so we can choose a suitable n for any $\epsilon > 0$ such that $v_i(x_n - a_i) < \epsilon$ for any i. $\square$

Corollary 23. $k_{\mathfrak{m}}$ is dense in k_∞^* under the embedding

$$i : k \hookrightarrow k_\infty, \quad x \mapsto (x_\sigma)_{\sigma \in \Sigma_o}$$

where Σ_0 is the set of ditinct embeddings up to conjuation, and $x_\sigma = \sigma(x)$ is the image of x under embedding σ.

Proof. Let $S = S_{\mathfrak{m}} \cup S_\infty$ be a set of absolute values on k, where $S_{\mathfrak{m}}$ is the set of non-archmedean absolute values given by prime ideals diving $\mathfrak{m}$, while S_∞ is the set of archemedean absolute values given by $v(x) = |\sigma(x)|$ for any $\sigma \in \Sigma_0$. Hence we choose $a_v = 1$ if $v \in S_{\mathfrak{m}}$, and $a_v \in k^*$ arbitrary if $v \in S_\infty$, apply the weak approximation theorem, we can get an element $x \in k$ such that $v(x-1) < 1$ for any $v \in S_{\mathfrak{m}}$, and $v(x - a_v) < \epsilon$ for any $v \in S_\infty$. The first condition implies that $x \in k_{\mathfrak{m}}$, while the second condition implies that $i(x)$ is close to $(a_v)_v$ in k_∞^*, hence we can conclude that $i(k_{\mathfrak{m}}) \subset i(k^*)$ dense, and then we can conclude that $k_{\mathfrak{m}}$ is dense in k_∞^* since $i(k)$ is dense in k_∞. $\square$

With enough preparation, we can define the character defined by Hecke (1920) in his original paper as following

Definition 24. Let $\mathfrak{m}$ be an ideal of $\mathcal{O}_k$, a **Größencharakter modulo** $\mathfrak{m}$ is a character $\chi : J(\mathfrak{m}) \to \mathbb{S}^1$ for which there eixsts a pair of (continous) characters

$$\chi_{\mathfrak{m}} : (\mathcal{O}_k/\mathfrak{m})^* \to \mathbb{S}^1, \quad \chi_\infty : k_\infty^* \to \mathbb{S}^1$$

such that

$$\chi((a)) = \chi_{\mathfrak{m}}(\bar{a}) \cdot \chi_\infty(i(a))$$

for any $a \in k(\mathfrak{m}) \cap \mathcal{O}_k$.

Remark 25. Here are several explanations:

- χ_m can be identifies as a character $f : k(\mathfrak{m}) \to \mathbb{S}^1$ as before, what we do here is to translate the relation before back to $\chi_{\mathfrak{m}}$.
- Review that $\chi((a)) = \chi_\infty(i(a)$ for any $a \in k_{\mathfrak{m}}$, the condition of continuity uniqeuly determines χ_∞ since embedding $k_{\mathfrak{m}}$ is dense in k_∞^*.
- Conversely, we can determine χ by the values on $P(\mathfrak{m})$, but it is not unqiue.

4.2 Ideal number system

In Hecke's papaer, he uses the notation of ideal numbers to make sum on ideals can be written as sum of elemenets, that is what we done above in Dedekind zeta function to to make the functional equation.

Definition 26. For a number field k, the **ideal number system** of k is a multiplicative abelian group $\hat{k}^*$ containing k^* such that the following diagram commutes and the rows are exact sequences

$$\begin{tikzcd}
1 \arrow[r] & \mathcal{O}_k^* \arrow[r, hook] \arrow[d, dash] & k^* \arrow[r] \arrow[d, hook] & P_k \arrow[r] \arrow[d, hook] & 1 \\
1 \arrow[r] & \mathcal{O}_k^* \arrow[r] & \hat{k}^* \arrow[r] & J_k \arrow[r] & 1
\end{tikzcd}$$

The construction of the system can be done as following:

By the theory of Minkowski, there exists a prime ideal $\mathfrak{p}$ in each class in class group J_k/P_k, so for any class we choose exactly one prime ideal $\mathfrak{p}$ as the representative, then there eixts a divisor d of class number h_k such that $\mathfrak{p}^d = (a)$ for some $a \in \mathcal{O}_k$, hence we define $\hat{p} \in \bar{k}^*$ as an ideal number of $\mathfrak{p}$ which statisfies $(\hat{p}) = \mathfrak{p}$, immediately $\hat{p}^d = ua$ for some unit u.

Then we can define $\hat{k}^*$ as a group generated by k^* and finite ideal numbers $\hat{p}$ of all classes (notice that in principal class, $\hat{p}$ is just in k^*). Now if we take any fractional ideals $\mathfrak{b}$, it will be in some class $[\mathfrak{p}]$, then there exists $x \in k^*$ such that

$$\mathfrak{b} = (x)\mathfrak{p} = (x)(\hat{p})$$

anyway some problem here ...

some properties of the system:

- By two exact sequences, we have $k^*/\mathcal{O}_k^* \cong P_k$ and $\hat{k^*}/\mathcal{O}_k^* \cong J_k$, hence by isomorphism theorem, we can conclude an isomorphism for class group

$$\hat{k^*}/k^* \cong J_k/P_k$$

- The norm of elements can be extends to ideal numbers as following:

$$\mathcal{N}(\hat{a}) = N((\hat{a}))$$

  i.e. make use of the absolute norm of ideals, clearly if we restrict the norm on k^*, it will be the algebraic norm $N_{k/\mathbb{Q}}$.

- Let $\hat{a} \in k^*$, we call it an ideal integer if $(\hat{a})$ is an integral ideal, and we define $\widehat{\mathcal{O}}$ as the set of all ideal integers, it is a semi-group (may not have a) under multiplication

4.3 Primitive character and Gauss sum

4.4 Local field

Some preparations about valuations are necessary here to claim some notations:

Definition 27. Let K be a field, **a vluation (or a absolute value)** on K is a map $v : K \to \mathbb{R}_{\geq 0}$ satisfying the following conditions:

1. $v(x) = 0 \iff x = 0$
2. $v(xy) = v(x)v(y)$
3. there exists a constant $C > 0$ such that $v(1 + x) \leq C$ whenever $v(x) \leq 1$.

Remark 28. The third condition is special here: if $C = 2$, it is equivalent to the triangle inequality $v(x+y) \leq v(x)+v(y)$, in this case we call v an archimedean valuation; if $C = 2$, then it is equivalent to the weak triangle inequality $v(x+y) \leq \max\{v(x), v(y)\}$, in this case we call v a non-archimedean valuation; If $v(x) = 1$ for any $x \neq 0$, then it is called a **trivial valuation**, usually we do not consider it.

Lemma 29. Any valuation v induces a topology on K with the open basis given by

$$B_r(a) = \{x \in K \mid v(a - b) < d\}$$

for any $a \in K$ and $d > 0$, and any two valuations v_1 and v_2 are equivalent (i.e. they induce the same topology) if there exists a constant $C > 0$ such that $v_1(x) = v_2(x)^C$ for any $x \in K$.

Remark 30. Notice the topology here is pre-metrizable by define a quasi-metric

$$d_v(x, y) = v(x - y)$$

it satisfies definition of metric except the inequality. In particular, if v is archimedean, then d_v is exactly a metric; If v is non-archimedean, then d_v is an ultrametric.

Let $\mathcal{V}$ be the set of all non-trivial valuations on K, and we define a equivalence relation on $\mathcal{V}$ by

$$v_1 \sim v_2 \iff v_1 \text{ is equivalent to } v_2$$

Then it induces a set $V(k) = \mathcal{V}/\sim$ of equivalence classes, and we call each class as **a place** of K, and we claim that in each calss, there exists a valuation v such that d_v is exactly a metric, if d_v is non-archemedean, we call v a prime place and denote it by $v \nmid \infty$; if d_v is archimedean, we call v an infinite place and denote it by $v \mid \infty$.

Theorem 31. Let k be a number field, then the $V(k)$ is determined by the following valuations:

- For any prime ideal $\mathfrak{p}$ of $\mathcal{O}_k$, there exists a discrete additive valuation on k^* by

$$v_{\mathfrak{p}}(x) = \max\{n \in \mathbb{Z} \mid \mathfrak{p}^n \text{ divise } (x)\}$$

it defines a non-archemedean aboslute value on k by

$$|x|_{\mathfrak{p}} = e^{-v_{\mathfrak{p}}(x)}$$

each prime ideal gives distinct prime places of k as above.

- For any embedding $\sigma : k \to \mathbb{C}$, it defines an aboslute value on k by

$$|x|_\sigma = |\sigma(x)|$$

where $|-|$ denotes the usual absolute value on $\mathbb{C}$. Distinct embeddings **up to conjuation** give all infinite places of k as above.

Remark 32. When $k = \mathbb{Q}$, the theorem is just Ostrowski's theorem, which states that any non-trivial valuation on $\mathbb{Q}$ is equivalent to either the p-adic valuation for some prime p, or the usual absolute value.

Remark 33. Any normed field attains an unique completion, in this case we denote the completion of $(k, |-|_v)$ by k_v, it is clearly that k_v is $\mathbb{Q}$ or $\mathbb{C}$ if v is an infinite place, in particular the minkowski's space is just the product of completions at infinite places

$$\prod_{v|\infty} k_v \cong k_\infty \cong k \otimes_{\mathbb{Q}} \mathbb{R}$$

with r the number of real embeddings and s the number of pairs of complex embeddings.

For prime places, each prime ideal lies on a unqiue prime number p, hence similarly we can conclude

$$\prod_{\mathfrak{p}|p} k_{\mathfrak{p}} \cong k \otimes_{\mathbb{Q}} \mathbb{Q}_p$$

4.5 Adele Ring

To define adele ring of a number field, we introduce the restricted direct product of toplogical spaces as following:

Lemma 34. Let $\{X_i\}_{i\in I}$ be a family of topological spaces, and $\{U_i\}_{i\in I}$ be a family of open subset of X_i, then the **the restricted direct product of** $\{X_i\}$ **with respect to** $\{U_i\}$ is defined as a subspace of direct product $\prod_{i\in I} X_i$ as following

$$\prod_{i\in I}' X_i := \{(x_i)_{i\in I} \in \prod_{i\in I} \mid x_i \in U_i \text{ for almost all } i \in I\}$$

it can be viewed as a topological space with the open basis given by

$$\mathcal{B} = \{\prod_{i\in I} V_i \mid V_i = U_i \text{ for almost all } i \in I\}$$

Lemma 35. If X_i is a locally compact group for any $i \in I$, and U_i is a compact open subgroup of X_i for any $i \in I$, then the restricted direct product $\prod_{i\in I}' X_i$ is a locally compact group under the restricted product.

Lemma 36. measure

Remark 37. If I is a finite index set, then whatever open subset U_i we choose, the restricted direct product is reduced to the direct product, i.e.

$$\prod_{i\in I}' X_i = \prod_{i\in I} X_i$$

Definition 38. Along the notation above, if k is a number field, then we define the finite adele ring of k as the restricted direct product of local fields at prime places

$$\mathbb{A}_{k,f} = \prod_{\mathfrak{p}\nmid\infty}' k_{\mathfrak{p}} \qquad \text{with respect to } \mathcal{O}_{\mathfrak{p}}$$

and the adele ring of k is the direct product of finite adele ring and the minkowski's space

$$\mathbb{A}_k = k_\infty \times \mathbb{A}_{k,f}$$

the addition and multiplication on $\mathbb{A}_k$ are defined componentwise.

Remark 39. Here are some quick descriptions of the adele ring:

- The adele ring is a locally compact. It is enough to proved that the finite adele ring is locally compact by lemma 35, we just set X_i by additive locally compact group $k_{\mathfrak{p}}$, and notice that local ring $\mathcal{O}_{\mathfrak{p}}$ is a compact open subgroup.
- The adele ring is a topological ring. It is enough to prove that the multiplication is continous

  $$m : \mathbb{A}_k \times \mathbb{A}_k \to \mathbb{A}_k, \quad ((x_i)_i, (y_i)_i) \mapsto (x_i \cdot y_i)$$

  by the universal property of product topology, it is continous if and only if each compoent is a topological ring.

- If $k = \mathbb{Q}$, we can conclude that
$$\mathbb{A}_{\mathbb{Q}} = \mathbb{R} \times \prod_p \mathbb{Q}_p$$
where p runs all prime numbers, hence generally for any number field k, we can conclude that
$$\mathbb{A}_k \cong k \otimes_{\mathbb{Q}} \mathbb{A}_{\mathbb{Q}}$$
is a isomorphism of topological rings, hence as a $\mathbb{A}_{\mathbb{Q}}$-module, $\mathbb{A}_k$ is free of rank $[k : \mathbb{Q}]$.
- It is natural to consider the digonal embedding
$$\Delta : k \to \mathbb{A}_k, \quad x \mapsto (x_v)_{v \in V(k)}$$
the image of Δ is exactly a subring of adele ring, with the subspace topology,we call it pincipal aldele, it is a discrete subgroup.

In claasical harmonic analysis, we usually consider groups $\mathbb{R}$, $\mathbb{Z}$ and $\mathbb{S}^1 \cong \mathbb{R}/\mathbb{Z}$, they are all locally compact abelian groups, and the dual group are $\mathbb{R}$, $\mathbb{S}^1$ and $\mathbb{Z}$ respectively, we can turn to the harmonic analysis in the context of adele ring, by the topological analogue of the groups.

Lemma 40. For any number field k, the adele ring $\mathbb{A}_k$ is a locally compact topological ring, and the principal adele k is a discrete subgroup of $\mathbb{A}_k$, and the quotient group $\mathbb{A}_k/k$ is compact.

4.6 Idele group

Definition 41. For any number field k, we define the (algebraic) idele group as the multiplicative group of units in the adele ring, that is
$$\mathbb{I}_k = \mathbb{A}_k^* = \prod_{v \in V(k)} k_v^*$$

As John Tate wrote in his thesis, we can do nothing signficant with idele until we consider the embedding of k.

The reason is a little delicated, in general, for a topological ring R, its subspace R^* is exactly a group, but not necessarily a topological group, since the the inverse map is not necessarily continous, **we do not need towards to other examples, adele ring is exactly such an example!**

Lemma 42. If $\mathbb{I}_k$ is viewed as a topological subspace of $\mathbb{A}_k$, then the inverse map
$$i : \mathbb{I}_k \to \mathbb{I}_k, \quad (a_v)_v \mapsto (a_v^{-1})_v$$
is not continous.

Proof. □

Remark 43. In general, if R is a topological ring such that R^* is a topological group under the subspace topology, then R is called an absolute tolopogical ring, a short note can be found in [1]

Hence we should consider another topology on idele group such that it can be a topological group.

Definition 44. The idele topology on $\mathbb{I}_k$ is defined as the restricted direct product with respect to $\mathcal{O}_\mathfrak{p}^*$, equivalently, it is the smallest topology such that the following map continous

$$j : \mathbb{I}_k \to \mathbb{A}_k \times \mathbb{A}_k, \quad x \mapsto (x, x^{-1})$$

Remark 45. Here are some quick notes:

- When it comes to the idele group, we always consider it with the idele topology, and then it is a locally compacttopological group since k_v^* is locally compact and $\mathcal{O}_k^*$ is a compact open subgroup for any place v.
- The digonal map $k^* \to \mathbb{I}_k, x \mapsto (x_v)_v$ naturally embeds k^*, and its image as a subgroup of $\mathbb{I}$, we call it as group principal ideles, it is a discrete subgroup of idele group.

  In fact, k^* is discrete in $\mathbb{A}_k$, and j is a continous injection, hence $j(k^*)$ is discrete implies that k^* is discrete in $\mathbb{I}_k$ under the idele topology.
- We define the content (magnitude) of an idele as a positive map

  $$c : \mathbb{I}_k \to \mathbb{R}_{>0}, \quad a = (a_v)_v \mapsto \prod_v \|a\|_v$$

  where v runs all places and $\| - \|_v$ is the normalized valuation on k_v, it defines a continous homomorphism with kernel denoted by $\mathbb{I}_k^1$.

It is better to handle $\mathbb{I}_k^1$ as a special subgroup of idele group, it can be viewed as the radical part of idele group when we consider the decomposition, here are the related properties:

Lemma 46. Along the notation above, we have the following properties:

1. I_k^1 is a closed subgroup of $\mathbb{I}_k$, so it is a locally compact group under the subspace topology.
2. The adele-subspace topology and idele-subspace topology on $\mathbb{I}_k^1$ are same.
3. There exists an isomoprhism $\mathbb{I}_k \cong \mathbb{I}_k^1 \times \mathbb{R}_{>0}$.
4. k^* can be identified as a subgroup of $\mathbb{I}_k^1$ under the digonal embedding, and the quotient group $\mathbb{I}_k^1/k^*$ is compact.

Proof. □

References

[1] Abolfazl Tarizadeh. *Some notes on topological rings and their groups of units*. 2025. arXiv: 2303.02634 [math.AC]. URL: https://arxiv.org/abs/2303.02634.